Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (34 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. The answer key is at the back.

Part A · Multiple choice

Choose one answer for each question.

A1 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO1

The general formula of an alkane is:

A. C_nH_{2n}

B. C_nH_{2n+2}

C. C_nH_{2n-2}

D. C_nH_n

A2 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO1

The IUPAC name of $(CH_3)_3CCH_2C(CH_3)_3$ is: (Ball 12.E)

A. 2,2,4,4-tetramethylpentane

B. 2,4-dimethylpentane

C. 1,1,3,3-tetramethylpentane

D. nonane

A3 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO1

Which compound is an isomer of hexane?

A. 2-methylpentane
B. cyclohexane
C. 2-methylhexane
D. hexene

A4 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO2

Which has the highest boiling point?

A. butane
B. pentane
C. 2-methylbutane
D. 2,2-dimethylpropane

A5 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO3

Incomplete combustion of an alkane produces:

A. only CO_2 and H_2O
B. CO, soot and H_2O
C. an alkene
D. a haloalkane

A6 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO3

In the free-radical chlorination of methane, the initiation step is:

A. $Cl_2 \rightarrow 2\ Cl\bullet$
B. $Cl\bullet + CH_4 \rightarrow \bullet CH_3 + HCl$
C. $\bullet CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\bullet$
D. $2\ \bullet CH_3 \rightarrow C_2H_6$

A7 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO4

The correct IUPAC name of $CH_3CH=CHCH_2CH_3$ is:

A. 3-pentene
B. 2-pentene
C. 2-pentyne
D. pentane

A8 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO4

Which can show cis-trans isomerism?

A. 1-butene
B. 2-methylpropene
C. 2-butene
D. ethene

A9 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO4

A chiral carbon is bonded to:

A. two identical groups
B. a double bond
C. four different groups
D. a benzene ring

A10 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO5

The major product of propene + HCl is:

A. 1-chloropropane
B. 2-chloropropane
C. 1,2-dichloropropane
D. propane

A11 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO5

Which reagent converts an alkene to an alkane?

A. H_2 with Ni

B. Br_2

C. H_2O with H_2SO_4

D. HNO_3

A12 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO5

Benzene typically undergoes:

A. addition with Br_2 at room temperature

B. substitution, keeping its aromatic ring

C. polymerisation

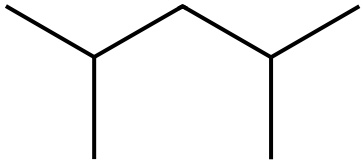
D. no reactions at all

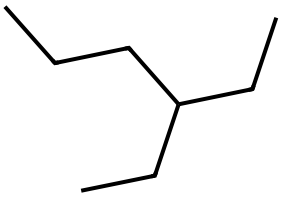
Part B · Structured questions

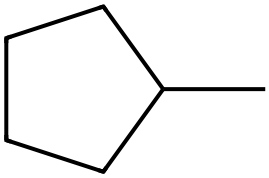
Write your answers on paper, then self-mark with the worked solution.

Question B1 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO1 8 marks, C3

(a) Draw the condensed structural formula of (i) 3-methylpentane, (ii) 2,2,5-trimethylhexane and (iii) 4-ethyl-3-methyloctane. (3 marks) (Ball 12.E) (b) Give the IUPAC name of each compound below. (3 marks)







(c) A student names a compound "2-ethylpentane". Explain the mistake and give the correct name. (2 marks)

Question B2 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO2 6 marks, C4

Pentane (bp 36 °C) and 2,2-dimethylpropane (bp 9.5 °C) have the same molecular formula. (a) State the molecular formula and the relationship between them. (2 marks) (b) Explain the difference in boiling points. (2 marks) (c) Explain why both are insoluble in water but float on it. (2 marks)

Question B3 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO3 7 marks, C3

(a) Write a balanced equation for the complete combustion of propane. (2 marks) (b) State one danger of incomplete combustion in a poorly ventilated room. (1 mark) (c) Methane reacts with chlorine in UV light. Write equations for the initiation step, the two propagation steps and one termination step. (4 marks)

Question B4 Topic: Saturated and Unsaturated Hydrocarbons · CH2-LO4 · 6 marks, C3

(a) Name: (i) $CH_3CH{=}C(CH_3)CH_2CH_3$ (ii) $CH{\equiv}CCH_2CH(CH_3)CH_3$ (2 marks) (b) Draw and name the cis and trans isomers of 2-pentene. (2 marks) (c) Explain why 1-pentene cannot show cis-trans isomerism. (1 mark) (d) Identify the chiral carbon in 2-butanol, $CH_3CH(OH)CH_2CH_3$, and explain why it is chiral. (1 mark)

Question B5 Topic: Saturated and Unsaturated Hydrocarbons · CH2-LO5 · 7 marks, C4

Starting from propene, $CH_2{=}CHCH_3$: (a) give the structure and name of the major product with (i) H_2/Ni, (ii) Br_2, (iii) HBr, (iv) H_2O/H_2SO_4. (4 marks) (b) State the rule used in (iii) and (iv). (1 mark) (c) Draw three repeating units of the polymer formed from propene and name it. (2 marks)

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Adapted from LibreTexts *Introduction to Organic Chemistry – Alkanes* 12.3, 12.6–12.10 and exercises 12.E (Sharpe Elles, from Ball, Hill & Scott), *The Basics of GOB Chemistry* (Ball, Hill & Scott) chapter 13 and *Fundamentals of General Organic and Biological Chemistry* 13.1–13.9, CC BY-NC-SA 3.0. Reaction mechanisms, Markovnikov items and pharmacy scenarios are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.

PHD125 Pharmaceutical Organic Chemistry · Saturated and Unsaturated Hydrocarbons

Tutorial: exam-style practice

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Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (34 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO1

The general formula of an alkane is:

A. C_nH_{2n}

B. C_nH_{2n+2}

C. C_nH_{2n-2}

D. C_nH_n

Saturated open chains: C_nH_{2n+2}.

A2 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO1

The IUPAC name of $(CH_3)_3CCH_2C(CH_3)_3$ is: (Ball 12.E)

A. 2,2,4,4-tetramethylpentane

B. 2,4-dimethylpentane

C. 1,1,3,3-tetramethylpentane

D. nonane

Five-carbon chain with two methyls on C2 and two on C4.

A3 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO1

Which compound is an isomer of hexane?

A. 2-methylpentane

B. cyclohexane

C. 2-methylhexane

D. hexene

2-Methylpentane is C_6H_{14}, like hexane. Cyclohexane and hexene are C_6H_{12}.

A4 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO2

Which has the highest boiling point?

A. butane

B. pentane

C. 2-methylbutane

D. 2,2-dimethylpropane

Pentane: unbranched, so largest contact area among the C_5 isomers, and bigger than butane.

A5 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO3

Incomplete combustion of an alkane produces:

A. only CO_2 and H_2O

B. CO, soot and H_2O

C. an alkene

D. a haloalkane

Limited O_2 gives carbon monoxide and soot.

A6 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO3

In the free-radical chlorination of methane, the initiation step is:

A. $Cl_2 \rightarrow 2\ Cl\bullet$

B. $Cl\bullet + CH_4 \rightarrow \bullet CH_3 + HCl$

C. $\bullet CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\bullet$

D. $2\ \bullet CH_3 \rightarrow C_2H_6$

UV light splits Cl_2 homolytically.

A7 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO4

The correct IUPAC name of $CH_3CH{=}CHCH_2CH_3$ is:

A. 3-pentene

B. 2-pentene

C. 2-pentyne

D. pentane

Number so the C=C gets the lower number.

A8 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO4

Which can show cis-trans isomerism?

A. 1-butene

B. 2-methylpropene

C. 2-butene

D. ethene

Only 2-butene has two different groups on each double-bond carbon.

A9 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO4

A chiral carbon is bonded to:

A. two identical groups

B. a double bond

C. four different groups

D. a benzene ring

Four different groups give two non-superimposable mirror images.

A10 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO5

The major product of propene + HCl is:

A. 1-chloropropane

B. 2-chloropropane

C. 1,2-dichloropropane

D. propane

Markovnikov: H to the CH_2 end, Cl to the middle carbon.

A11 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO5

Which reagent converts an alkene to an alkane?

A. H_2 with Ni

B. Br_2

C. H_2O with H_2SO_4

D. HNO_3

Hydrogenation.

A12 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO5

Benzene typically undergoes:

A. addition with Br_2 at room temperature

B. substitution, keeping its aromatic ring

C. polymerisation

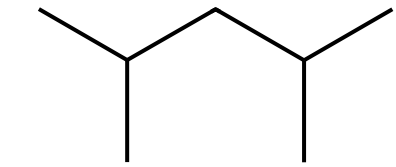
D. no reactions at all

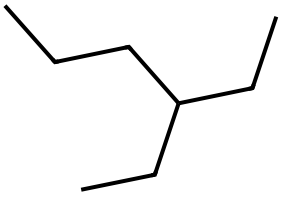
Its delocalised ring is stable, so it substitutes rather than adds.

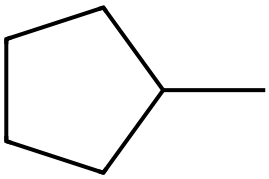
Part B · Structured questions

Question B1 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO1 8 marks, C3

(a) Draw the condensed structural formula of (i) 3-methylpentane, (ii) 2,2,5-trimethylhexane and (iii) 4-ethyl-3-methyloctane. (3 marks) (Ball 12.E) (b) Give the IUPAC name of each compound below. (3 marks)







(c) A student names a compound "2-ethylpentane". Explain the mistake and give the correct name. (2 marks)

(a)(i) $CH_3CH_2CH(CH_3)CH_2CH_3$ (ii) $(CH_3)_3CCH_2CH_2CH(CH_3)_2$ (iii) $CH_3CH_2CH(CH_3)CH(CH_2CH_3)CH_2CH_2CH_2CH_3$. (b) 2,4-dimethylpentane; 3-ethylhexane; methylcyclopentane. (c) The ethyl group on C2 extends the longest chain to six carbons, so the parent is hexane: the correct name is **3-methylhexane**.

Step	Marking criteria	Marks
1	(a)(i) correct	1
2	(a)(ii) correct	1
3	(a)(iii) correct	1
4	(b) 2,4-dimethylpentane	1
5	(b) 3-ethylhexane	1
6	(b) methylcyclopentane	1
7	(c) Mistake: longest chain not used (ethyl extends it)	1

8	(c) 3-methylhexane	1
	Total	**8**

Question B2 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO2 6 marks, C4

Pentane (bp 36 °C) and 2,2-dimethylpropane (bp 9.5 °C) have the same molecular formula. (a) State the molecular formula and the relationship between them. (2 marks) (b) Explain the difference in boiling points. (2 marks) (c) Explain why both are insoluble in water but float on it. (2 marks)

(a) C_5H_{12}; they are structural isomers. (b) Both have only dispersion forces. Pentane is unbranched, with a larger surface area and more contact between molecules, so its dispersion forces are stronger and more energy is needed to separate the molecules. 2,2-Dimethylpropane is compact and nearly spherical, so it has less contact. (c) Alkanes are nonpolar and cannot form hydrogen bonds with water, so they do not dissolve; their densities (about 0.6 g/mL) are lower than water's (1.0 g/mL), so they float.

Step	Marking criteria	Marks
1	C_5H_{12}	1
2	Structural isomers	1
3	Dispersion forces only	1
4	Branching reduces surface contact / weaker forces	1
5	Nonpolar, so insoluble	1
6	Less dense than water	1
	Total	**6**

Question B3 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO3 7 marks, C3

(a) Write a balanced equation for the complete combustion of propane. (2 marks)
(b) State one danger of incomplete combustion in a poorly ventilated room. (1 mark)
(c) Methane reacts with chlorine in UV light. Write equations for the initiation step, the two propagation steps and one termination step. (4 marks)

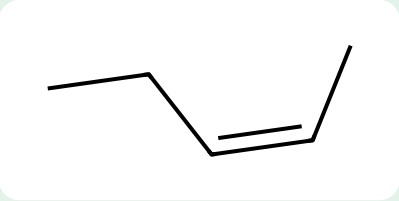
(a) $C_3H_8 + 5\ O_2 \rightarrow 3\ CO_2 + 4\ H_2O$. (b) Carbon monoxide forms; it binds to haemoglobin and stops oxygen transport (CO poisoning). (c) Initiation: $Cl_2 \rightarrow 2\ Cl\bullet$ (UV). Propagation: $Cl\bullet + CH_4 \rightarrow \bullet CH_3 + HCl$; $\bullet CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\bullet$. Termination (any one): $Cl\bullet + Cl\bullet \rightarrow Cl_2$; $\bullet CH_3 + Cl\bullet \rightarrow CH_3Cl$; $\bullet CH_3 + \bullet CH_3 \rightarrow C_2H_6$.

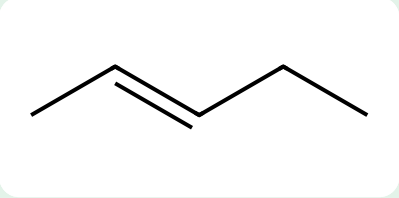
Step	Marking criteria	Marks
1	Correct formulas of reactants and products	1
2	Balanced (1 : 5 : 3 : 4)	1
3	CO poisoning stated	1
4	Initiation	1
5	Propagation 1	1
6	Propagation 2	1
7	Any valid termination	1
	Total	**7**

Question B4 Topic: Saturated and Unsaturated Hydrocarbons CH2-LO4 6 marks, C3

(a) Name: (i) $CH_3CH=C(CH_3)CH_2CH_3$ (ii) $CH{\equiv}CCH_2CH(CH_3)CH_3$ (2 marks) (b) Draw and name the cis and trans isomers of 2-pentene. (2 marks) (c) Explain why 1-pentene cannot show cis-trans isomerism. (1 mark) (d) Identify the chiral carbon in 2-butanol, $CH_3CH(OH)CH_2CH_3$, and explain why it is chiral. (1 mark)

(a)(i) 3-methyl-2-pentene (ii) 4-methyl-1-pentyne. (b) cis-2-pentene: CH_3 and CH_2CH_3 on the same side; trans-2-pentene: on opposite sides.





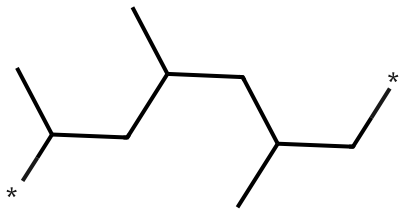
(c) C1 of 1-pentene carries two H atoms (two identical groups), so swapping sides gives the same molecule. (d) C2: it carries four different groups: H, OH, CH_3 and CH_2CH_3.

Step	Marking criteria	Marks
1	3-methyl-2-pentene	1
2	4-methyl-1-pentyne	1
3	cis isomer drawn and named	1
4	trans isomer drawn and named	1
5	Two identical H on C1	1
6	C2 with four different groups listed	1
	Total	**6**

Question B5 Topic: Saturated and Unsaturated Hydrocarbons | CH2-LO5 | 7 marks, C4

Starting from propene, $CH_2{=}CHCH_3$: (a) give the structure and name of the major product with (i) H_2/Ni, (ii) Br_2, (iii) HBr, (iv) H_2O/H_2SO_4. (4 marks) (b) State the rule used in (iii) and (iv). (1 mark) (c) Draw three repeating units of the polymer formed from propene and name it. (2 marks)

(a)(i) $CH_3CH_2CH_3$, propane (ii) $CH_2BrCHBrCH_3$, 1,2-dibromopropane (iii) $CH_3CHBrCH_3$, 2-bromopropane (iv) $CH_3CH(OH)CH_3$, 2-propanol. (b) Markovnikov's rule: H adds to the double-bond carbon that already has more H atoms. (c) –$CH_2CH(CH_3)CH_2CH(CH_3)CH_2CH(CH_3)$–, polypropylene.



Step	Marking criteria	Marks
1	(i) propane	1
2	(ii) 1,2-dibromopropane	1
3	(iii) 2-bromopropane	1
4	(iv) 2-propanol	1
5	Markovnikov's rule stated	1
6	Correct repeating unit (CH_3 on alternate C)	1
7	Named polypropylene	1
	Total	**7**

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